



## Srushti Nerkar

Roll No.: 2022UCP1661

Bachelors in Technology

Computer Science and Engineering

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GitHub Profile

LinkedIn Profile

Portfolio

## EDUCATION

### •Malaviya National Institute of Technology, Jaipur

2026

Computer Science and Engineering

### •Unnati High School

2022

Higher Secondary School Certificate, Maharashtra

## TECHNICAL SKILLS AND INTERESTS

**Languages:** C++, Python, Java, JavaScript, SQL

**Web Technologies:** React.js, Node.js, Express.js, TailwindCSS, HTML5/CSS3

**Machine Learning & AI:** Machine Learning, Deep Learning, Natural Language Processing (NLP), RAG, LLM Integration

**System Design & Architecture:** HLD (Microservices, Scalability), LLD (Design Patterns, SOLID Principles)

**Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis

**DevOps & Tools:** Docker, Git, GitHub, VS Code, Linux/Unix (Shell Scripting)

**Networking & OS:** TCP/IP, Linux Networking Commands

## EXPERIENCE

### •Birdvision Consulting Pvt. Ltd.

May 2025 - July 2025

Software Development Intern

Pune

- Engineered scalable backend microservices using Flask and integrated WhatsApp Business APIs for 10K+ user interactions.
- Designed RESTful microservices for messaging, routing, and status handling.
- Improved reliability by optimizing API error handling and response processing.

## PROJECTS

### •PRISM: Progressive Routing for Intrusion Detection via Spiking Multi-layer Networks

May 2026

Python, PyTorch, Scikit-learn, Spiking Neural Networks (SNN), Isolation Forest, Ensemble Learning

- Co-engineered PRISM, a hierarchical IDS cascading an Isolation Forest global pre-filter with a 3-layer Parametric LIF Spiking Neural Network (SNN) and a 4-model stacked ensemble.
- Resolved threshold contradictions in hierarchical IDS by decoupling anomaly detection from classification, intercepting zero-day threats via feature-space signatures before misrouting by the binary SNN classifier.
- Achieved 95.59% overall accuracy and a 0.758 macro F1-score on 377,698 test flows (CICIDS2017), securing a 60.3% overall zero-day detection rate at a low 3.39% false positive rate.

### •Smart Sentry: Real-Time Industrial Anomaly Detection Pipeline | GitHub Link

Baker Hughes Hackathon

Python, Scikit-learn, Pandas, Kafka, Redis, PostgreSQL, Streamlit, Docker

Oct 2025

- Engineered a phase-aware anomaly detection system to minimize false positives during transient operational states (startup/shutdown).
- Architected a low-latency, event-driven data pipeline using **Kafka** for stream ingestion and **Redis** for real-time state management, facilitating high-throughput processing for critical failure detection.
- Achieved 98% overall accuracy and a 0.71 F1-Score on the anomaly class, effectively isolating critical failures while maintaining robust performance under peak data load.

### •Aspect-Based Sentiment Analysis (ABSA) with Contextual Opinion Mining | GitHub Link

Dec 2025

Python, BERT, PyTorch, NLP, spaCy

- Engineered a **hybrid ABSA pipeline** integrating **fine-tuned BERT models** with rule-based heuristics to resolve granularity gaps in complex, multi-aspect sentiment analysis.
- Implemented Contrastive Clause Resolution and local context window mechanisms to identify explicit Opinion Phrases, providing verifiable, human-readable justification for model inferences.
- Achieved 92.1% accuracy and 0.90 Macro F1-Score on validation datasets, demonstrating model robustness and high classification performance across diverse sentiment classes.

## CODING PROFILE

–CodeForces (Max Rating - 1299, Pupil) Global Rank 2259 in Round 1021(Div 2) | Link

–LeetCode (Max Rating - 1834, Top 6.36%) Global Rank 720 in Weekly Contest 472 | Link | Link

–Geeks for Geeks (Max Rating - 1756, 3 star) Global Rank 210 in Weekly 189 | Link

–CodeChef (Max Rating - 1482, 2 star) Global Rank 401 in Starters 173 | Link

## ACHIEVEMENTS

–Finalist at Baker Hughes Hackathon 2025

2025

–Granted 3 Indian patents with 2 more pending for technological innovations

–Invited as speaker at TEDx MITAOE

2020

–Invited to speak at the International Women's Day celebration organized by WTM and GDG Pune

2021

–Featured in United Nations film for Clean Water Issue

2020